

GNN-Based BERT for Understanding Context from Music

Humaira Anzum Neha
ID: 1000060159

Department of Computer Science and Engineering
BRAC University

Sadman Sakib
ID: 1000060158

Department of Computer Science and Engineering
BRAC University

Abstract—Music carries context at several levels, from genre and mood to instrumentation, structure, and natural-language descriptions, making it difficult to understand from a single modality alone. This work investigates how semantic and structural representations can complement each other through a GNN-BERT framework for music context understanding. Following a four-task progression, we first fine-tune BERT for caption-based multi-label tagging, then model audio segments as graphs using GraphSAGE, combine graph and text representations through cross-attention for genre and mood prediction, and finally align audio graphs with natural-language captions through contrastive learning. Experiments on MusicCaps, GTZAN, and MagnaTagATune show that text provides a strong semantic signal, while the value of graph structure depends on the prediction target and graph construction. In the multimodal experiments, combining audio and text increased AUC-PR from 0.2464 for the audio-only representation to 0.3013, while controlled comparison with a no-graph MLP showed that message passing consistently benefited genre prediction but not mood prediction. Per-tag threshold tuning provided a further approximately +0.012 improvement in Macro-F1 on MagnaTagATune. Contrastive GNN-BERT training additionally enabled bidirectional audio-caption retrieval and zero-shot tag prediction without direct tag supervision. Overall, the results show that combining language semantics with structured audio representations can improve music context understanding, while also revealing an important practical lesson: adding graph structure is useful only when the relationships encoded by the graph carry information relevant to the target.

Index Terms—music information retrieval, graph neural networks, BERT, multi-label classification, contrastive learning, ablation study

I. INTRODUCTION

Music understanding has become an important research area in multimedia and machine learning due to its wide range of applications, including music recommendation, content retrieval, playlist generation, and automatic music annotation. Unlike traditional classification problems, music contains complex semantic information. A single music clip may simultaneously convey multiple concepts, such as genre, mood, instrumentation, and contextual descriptions. Therefore, developing models capable of understanding music from multiple perspectives remains a challenging problem.

Early approaches for automatic music analysis primarily relied on handcrafted audio features, including spectral descriptors, chroma features, and Mel-frequency cepstral coefficients (MFCCs), followed by conventional machine learning classifiers [1]. Although these methods achieved reasonable

performance for specific tasks, they often struggled to capture the high-level semantic meaning of music. Recent advances in deep learning have enabled end-to-end learning from raw audio representations, allowing models to discover complex patterns directly from data, and convolutional architectures over log-mel spectrograms are now the established approach to music auto-tagging [2], [3]. However, many existing approaches still treat audio signals as independent feature vectors and fail to exploit the relational structure present within music segments.

Graph neural networks (GNNs) provide a promising solution for modeling structured data by learning from relationships among interconnected elements [4], [5]. In music analysis, an audio track can naturally be represented as a graph where nodes correspond to temporal audio segments and edges represent relationships such as temporal continuity or acoustic similarity. By propagating information across neighboring segments, GNNs can capture global contextual patterns within a track, and recent work has applied this idea successfully to audio tagging and genre classification [6], [7].

Pretrained language models such as BERT [8] have demonstrated strong capability in learning semantic representations from natural language. When applied to music descriptions, BERT can capture meaningful relationships between textual descriptions and high-level music concepts. Nevertheless, text-based models lack direct access to acoustic information and may fail when descriptions are incomplete or ambiguous. This motivates the development of multimodal approaches that combine complementary information from both audio and language representations.

In this project, we investigate a multimodal framework that integrates graph-based audio representation learning with transformer-based language modeling for music context understanding. The proposed approach utilizes graph neural networks to learn structural representations from audio segments and BERT-based models to encode natural-language descriptions. These complementary representations are fused to perform multi-label prediction of music-related concepts, including genre and mood categories. Furthermore, we explore cross-modal representation learning through contrastive alignment [9], [10], where audio graphs and textual descriptions are projected into a shared embedding space to enable cross-modal retrieval.

The main contributions of this work are summarized as follows:

- We developed a graph-based representation of music tracks by modeling temporal audio segments as interconnected nodes and applying graph neural networks for audio representation learning.
- We investigate BERT-based semantic representation learning for music descriptions and formulate multi-label music context prediction using genre and mood annotations.
- We explore a multimodal fusion framework that combines graph-based audio embeddings and language-based embeddings to improve music understanding.
- We explore contrastive audio–text alignment to learn a shared embedding space between music representations and natural-language descriptions, enabling cross-modal retrieval.

II. RELATED WORK

A. Content-Based Music Auto-Tagging

Automatic music tagging is conventionally posed as multi-label binary classification over a fixed vocabulary, and the dominant approach applies convolutional networks to time–frequency representations. Tzanetakis and Cook [1] established the GTZAN benchmark and the single-label genre formulation that Task 2 of this work follows, while Law et al. [11] introduced MagnaTagATune, whose 188-tag vocabulary was collected through a two-player online game and remains a standard multi-label benchmark. Because published results are reported under inconsistent splits and preprocessing, Won et al. [12] re-evaluated the major CNN architectures on common splits and released reference ROC-AUC and PR-AUC figures.

B. Graph Neural Networks for Audio

Graph formulations of audio have gained traction because convolutional receptive fields remain local while transformers impose a rigid patch grid on what are often irregular acoustic objects. Singh et al. [6] treat the spectrogram itself as a graph and learn cross-correlations between class embeddings and spectrogram regions, and their later work [13] extends this with higher-order neighbourhood information, outperforming transformer baselines with substantially fewer parameters. Shrivastava and Bhattacharyya [7] apply graph representation learning to audio and music genre classification using similarity relations between segments, which is the construction closest to ours.

C. Language Models and Cross-Modal Alignment

BERT [8] supplies the frozen text encoder used in Tasks 3 and 4. On the cross-modal side, CLIP [9] established the dual-encoder recipe we follow: two towers trained with a symmetric InfoNCE objective [10] over in-batch negatives, followed by zero-shot classification obtained by embedding class names as text. We adopt this protocol directly for zero-shot tagging in Task 4, with the caveat that CLIP ensembles many prompt

TABLE I
DATASETS, TASKS AND PREPROCESSING. MUSICCAPS IS USED TWICE: TEXT-ONLY TAG CLASSIFICATION IN TASK 1, WHERE NO AUDIO IS REQUIRED, AND CROSS-MODAL ALIGNMENT IN TASK 4, WHERE IT IS SEGMENTED LIKE THE OTHER CORPORA.

Dataset	Task	Clips	Clip	Seg.	Fr.	Labels
MusicCaps (text)	1	5,501	–	–	–	65 multi
GTZAN	2	999	30 s	5 s	216	10 single
MagnaTagATune	3	17,414	29.1 s	5 s	216	53 multi
MusicCaps (audio)	4	5,299	10 s	2 s	87	55 multi

templates whereas we use one. MusicCaps [14] provides the paired (audio, expert caption) data that makes the contrastive setup viable, which is why we adopt it for Task 4.

III. DATASETS AND PREPROCESSING

We used different datasets for the four tasks according to their respective objectives. MusicCaps was used for the text-based Task 1 and the cross-modal Task 4, GTZAN was used for the audio-only GNN task, and MagnaTagATune was used for the multimodal GNN–BERT fusion task. Table I summarizes the datasets, task assignments, clip and segment configurations, feature dimensions, and label spaces used in our experiments. The following subsections describe the datasets and preprocessing procedures for each task.

A. Task 1: MusicCaps Caption-to-Tag Prediction

Task 1 used the MusicCaps dataset, which provides short music clips accompanied by natural-language descriptions. The input to the model was the textual caption associated with each clip, while the output consisted of curated music-related tags derived from the predefined aspect list. A total of 5,501 clips were used, with 65 multi-label tags. The resulting sequences were padded or truncated to a fixed maximum length before being provided to the pretrained BERT encoder.

The curated tag space was restricted to the selected semantic categories used in the experiment. Tags with fewer than 45 positive examples were excluded to avoid unreliable evaluation for extremely rare labels. The resulting problem was formulated as multi-label prediction, where each clip could be associated with multiple tags simultaneously. Label-wise stratification was used during splitting to preserve the distribution of the multi-label targets across the dataset splits.

B. Task 2: GTZAN Audio Graph Classification

Task 2 used the GTZAN dataset, which contains 30-second music tracks from ten genre categories. A total of 999 clips were used. Each audio track was resampled to 22,050 Hz and divided into non-overlapping 5-second segments. Thus, each track was represented as a sequence of fixed-length audio segments.

For each segment, audio features were extracted to provide the initial node representation for the graph neural network. Chroma and Mel-frequency cepstral coefficient (MFCC) representations were used to characterize the harmonic and timbral

properties of the audio segments. The resulting feature representation had 216 dimensions. The extracted features were normalized before graph construction. Each 5-second segment was represented as a graph node, while edges captured relationships between segments, including temporal adjacency and acoustic similarity based on the extracted features. The resulting graph therefore represented both the sequential structure of the track and relationships between acoustically similar segments. Label-wise stratification was used to maintain comparable genre distributions across the dataset splits.

C. Task 3: MagnaTagATune GNN-BERT Fusion

Task 3 used the MagnaTagATune dataset for multi-context music understanding. A total of 17,414 clips were used, with an average clip duration of 29.1 s. Each audio clip was resampled to 22,050 Hz and divided into 5-second segments. The audio segments were used to construct a graph representation in which each segment corresponded to a node. The extracted audio representation contained 216 features per segment.

In parallel, textual input was constructed from the meta-data associated with each track. The instrument tags were concatenated to form the textual input provided to BERT. The prediction target was formed from the genre and mood tags, resulting in a multi-label classification problem with 53 target labels. Tags with fewer than 45 positive examples were excluded from the target label space to reduce the instability associated with extremely rare labels. Label-wise stratification was used when forming the dataset splits.

D. Task 4: MusicCaps Cross-Modal Alignment

Task 4 used the MusicCaps dataset to investigate cross-modal alignment between audio and natural-language descriptions. A total of 5,299 audio clips were used. Each MusicCaps audio clip was resampled to 22,050 Hz and divided into 2-second segments for graph construction, with 87-dimensional audio features extracted for each segment. The resulting audio segments formed the nodes of the audio graph, with graph connections representing relationships between segments. The corresponding MusicCaps caption was retained as the natural-language representation of the same audio clip. The task used 55 multi-label tags for the associated music context.

The MusicCaps data were divided into training, validation, and test subsets using a label-wise stratification to maintain a consistent distribution of the available tag information across the splits. The test subset was kept isolated during model development and was used only for final cross-modal retrieval evaluation.

E. Leakage Control

To reduce the possibility of information leakage, the training, validation, and test subsets were kept disjoint at the dataset-unit level. Splitting was performed before model training, and label-wise stratification was used to preserve the distribution of the target labels across the subsets. All feature extraction and normalization procedures used for model development were performed without using test-set information.

Consequently, the test data remained unseen during training and model selection.

IV. METHOD

A. Audio Feature Extraction

Each audio clip was resampled to 22,050 Hz and divided into fixed-length 5-second segments. For each segment, MFCC and chroma features were extracted and pooled over time to obtain a fixed-dimensional representation v_i . Feature standardisation was performed using statistics computed from the training data.

B. Graph Construction

Each clip becomes a graph $G = (V, E)$ whose nodes are time segments. Node features are the standardised pooled vectors. Edges comprise a temporal chain $i \leftrightarrow i+1$ plus similarity edges between non-adjacent segments, with edge_attr = [is_temporal, similarity].

Similarity is cosine distance on *track-centred* vectors:

$$\tilde{v}_i = v_i - \frac{1}{|V|} \sum_j v_j, \quad s_{ij} = \frac{\tilde{v}_i^\top \tilde{v}_j}{\|\tilde{v}_i\| \|\tilde{v}_j\|}. \quad (1)$$

Centring is necessary rather than cosmetic: pooled chroma and MFCC-mean features are largely non-negative and dominated by a clip-level offset, so raw cosine similarity approaches 1.0 for every pair. Centring reframes the question as how each segment deviates from what is typical for that clip.

A consequence is that centred vectors sum to zero, so the mean pairwise cosine is pinned at $-1/(n-1)$ independently of the audio: -0.20 at six segments and -0.25 at five. Section VI uses this to explain the observed zero-edge rates. We compare two edge policies: a fixed threshold $\tau = 0.3$ and a top- k rule retaining the k most similar non-adjacent pairs per clip.

C. GraphSAGE Encoder

For each graph, node representations were updated using GraphSAGE. At layer l , a node aggregates the mean representation of its neighbors and combines it with its own representation:

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \text{CONCAT} \left(h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)} \right) \right). \quad (2)$$

After the final layer, mean pooling produces a track-level audio representation:

$$g = \frac{1}{|V|} \sum_{i \in V} h_i^{(L)}. \quad (3)$$

For the audio-only classification setting, the graph representation is passed to a prediction layer:

$$\hat{y} = \sigma(W_g g + b_g). \quad (4)$$

D. BERT Text Encoding and Tag Classification

For text-based and multimodal tasks, textual inputs were tokenized using the BERT tokenizer and padded or truncated to a fixed sequence length. BERT produces contextual token representations:

$$H_{\text{text}} = \text{BERT}(X_{\text{text}}). \quad (5)$$

The contextual representation of the [CLS] token was used as the global text representation.

For Task 1, we fine-tune bert-base-uncased end to end with a multi-label classification head using MusicCaps captions, which were lower-cased and whitespace-normalised with a maximum sequence length of 256. Targets comprise curated tags from the *genre*, *mood*, and *tempo* categories occurring in at least 45 clips, resulting in 65 labels. Training uses BCE-WithLogitsLoss with inverse-frequency pos_weight capped at 10. The data were split into training, validation, and test sets using iterative multi-label stratification with an 80/10/10 ratio. AdamW was used with a learning rate of 2×10^{-5} , weight decay of 10^{-5} , a 10% warmup ratio, and batch size 32. Early stopping was applied after four consecutive epochs without an improvement of at least 10^{-3} in validation loss; training converged in 26 epochs.

Unlike Tasks 3 and 4, BERT was fully fine-tuned for Task 1 because its inputs are genuine free-text captions with substantial linguistic variation. For the fusion experiments, the lower BERT layers were frozen and only the final transformer layers and task-specific prediction layers were fine-tuned.

E. Cross-Modal Fusion

For Task 3, the graph and textual representations were combined using cross-attention rather than direct concatenation. The graph-level audio representation acts as the query, while the BERT token representations provide keys and values:

$$Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad V = H_{\text{text}}W_V. \quad (6)$$

The attention weights are computed as

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad (7)$$

and the attended textual representation is

$$c = AV. \quad (8)$$

The attended representation is combined with the original graph representation:

$$z = \text{CONCAT}(g, c). \quad (9)$$

The fused representation is then used for multi-label prediction of genre and mood tags:

$$\hat{y} = \sigma(W_z z + b_z). \quad (10)$$

The model was trained using binary cross-entropy with logits, with class weighting used to account for label imbalance.

F. No-Graph Control

To determine whether graph message passing provides benefits beyond simple feature aggregation, we used a no-graph control model. The control receives the mean of the input node features directly,

$$\bar{x} = \frac{1}{|V|} \sum_i x_i, \quad (11)$$

and processes it using an MLP matched to the GNN in depth, width, dropout, optimizer, loss, and evaluation protocol. Thus, the primary architectural difference between the two models is the presence of graph message passing.

G. Contrastive Audio-Text Alignment

For Task 4, each audio graph was paired with its corresponding MusicCaps caption. The GNN and BERT encoders produce graph and text representations, which are projected and normalized before contrastive learning:

$$g_i = \text{Normalize}(f_g(G_i)), \quad t_i = \text{Normalize}(f_t(X_i)). \quad (12)$$

For a batch of N paired examples, the cross-modal similarity is

$$s_{ij} = g_i^\top t_j. \quad (13)$$

We use a symmetric InfoNCE objective with a learnable temperature τ_{temp} :

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2N} \sum_i \left[\log \frac{\exp(s_{ii}/\tau_{\text{temp}})}{\sum_j \exp(s_{ij}/\tau_{\text{temp}})} + \log \frac{\exp(s_{ii}/\tau_{\text{temp}})}{\sum_j \exp(s_{ji}/\tau_{\text{temp}})} \right]. \quad (14)$$

The first term aligns each audio graph with its corresponding caption, while the second performs the reverse caption-to-audio alignment. This symmetric objective is consistent with the bidirectional retrieval evaluation used in Task 4.

V. EXPERIMENTAL SETUP

All models use BCEWithLogitsLoss with inverse-frequency pos_weight capped at 10, AdamW ($\eta = 10^{-3}$, weight decay 10^{-5}), batch size 32, and early stopping on validation AUC-PR. Decision thresholds are tuned *per tag on the validation split only* and applied frozen to test.

VI. RESULTS

A. Task 1: BERT Tag Classification

Table II breaks the result down by tag category. Performance is highest on tempo (0.6763 AUC-PR over three tags) and lowest on mood (0.6112 over twenty-two), with genre between them an ordering that tracks how explicitly each property tends to be stated in a written caption. Overall Micro-F1 is 0.6394 (precision 0.6007, recall 0.6834), and training converged in 26 epochs to a best validation loss of 0.3221.

TABLE II

TASK 1: BERT MULTI-LABEL TAGGING FROM MUSICCAPS CAPTIONS. 65 TAGS, 504 TEST CLIPS, ALL TAGS WITH TEST EXAMPLES.

Category	Tags	Macro-F1	AUC-PR
Genre	40	0.6139	0.6631
Mood	22	0.5730	0.6112
Tempo	3	0.6464	0.6763
All	65	0.6016	0.6462

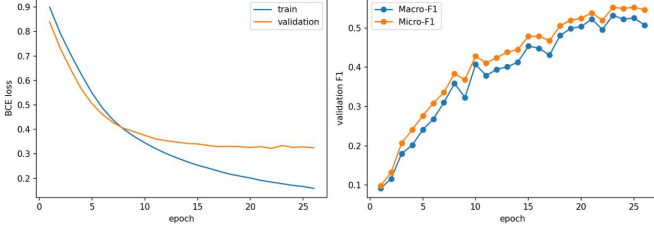


Fig. 1. Task 1: training and validation BCE loss (left) and validation Macro-F1 and Micro-F1 (right) against training epoch.

a) Why this number is high, and what it does not mean:

Task 1 substantially outperforms every audio-based model in this report (Section VI). The explanation is not that BERT is a better music model but that the task is close to circular: the curated tags were extracted from each clip’s aspect_list, written by the same annotator in the same listening pass as the caption that BERT reads. The specification itself designates this a “caption $\rightarrow$ tag proxy task”. Recovering tags from a paraphrase of themselves is a much easier problem than inferring them from audio, and Task 1 should be read as an upper reference point rather than as a competing system.

The comparison with the audio models isolates two effects at once. Predicting the same tag families from captions reaches 0.6462 AUC-PR; predicting them from the audio graph on the same corpus reaches 0.1502 supervised and 0.0535 zero-shot, against a label-prior baseline of 0.0462. That $4.3\times$ gap bounds how much of the label information is recoverable from a 26-dimensional pooled MFCC summary versus from language that already describes the answer, and it is the clearest statement in this work of the limitation discussed in Section VIII.

Figure 1 shows the training curves. Validation loss falls to approximately 0.32 by epoch 15 and then flattens while training loss continues to decline, so the model is mildly overfit from that point; both F1 measures continue to improve slowly to epoch 26, where early stopping on validation loss halted training. Micro-F1 leads Macro-F1 by a roughly constant margin throughout, which is the expected signature of a long-tailed tag distribution: the frequent tags that dominate Micro-F1 are learned first.

B. Task 2: GNN on Music Structure Graphs (GTZAN)

Table III reports all three splits, and Table IV the feature ablation. Test Macro-F1 of 0.6024 is $33\times$ the majority baseline,

TABLE III

TASK 2: GTZAN GENRE CLASSIFICATION, MFCC-ONLY GRAPH SAGE ($\tau = 0.3$, 3 LAYERS, 64 HIDDEN). ALL THREE SPLITS ARE SHOWN BECAUSE THE GAP BETWEEN THEM IS ITSELF INFORMATIVE.

Split	Accuracy	Macro-F1	Micro-F1	AUC-PR
Train	0.7840	0.7710	0.7840	0.8866
Validation	0.7200	0.7023	0.7200	0.7701
Test	0.6333	0.6024	0.6333	0.7164
Majority baseline	0.1000	0.0182	0.1000	0.1000

TABLE IV

TASK 2 FEATURE ABLATION, VALIDATION MACRO-F1. GRAPH SAGE THROUGHOUT.

Configuration	Features	Macro-F1	Micro-F1
Chroma-only	24-d	0.4682	0.4600
MFCC-only	26-d	0.7023	0.7200
Concat (early fusion)	50-d	0.6641	0.6733
Two-branch (late)	24+26-d	0.6880	0.6933
CNN mel-spectrogram [†]	128 $\times$ 216	0.7470	0.7508

and test AUC-PR of 0.7164 is the more stable figure since it does not depend on the argmax decision.

a) On reporting test rather than validation: The validation split was used for both early stopping and checkpoint selection, so validation figures are optimistically biased by construction; earlier validation-only reports of this configuration gave 0.7363. The train–validation gap is modest (0.069 Macro-F1), indicating the model is not severely overfit. With 150 test tracks over 10 classes 15 per class the standard error on accuracy is 0.039, and a single additional error per class moves accuracy by 0.067. Differences below roughly 0.08 Macro-F1 on this corpus are therefore not resolvable, which is why the chroma/MFCC gap (0.234, about six standard errors) is treated as real while smaller ones are not.

b) On the choice of backbone: GraphSAGE encoder is implemented in Task 2.

c) Chroma dilutes rather than complements: Chroma-only reaches $4.7\times$ chance but sits 0.234 below MFCC-only. Chroma collapses the spectrum onto twelve pitch classes and deliberately discards timbre, which is what genre labels largely track; pitch-class content also varies more within a genre than between genres. Critically, *both* ways of combining the modalities score below MFCC alone. Concatenation nearly doubles the input dimension and the two-branch model nearly doubles the encoder, and with only 699 training graphs that extra capacity is spent representing a feature that scores 0.4682 on its own.

d) Comparison with CNN Baseline: Figure 2 places the side by side comparison of CNN and GNN validation score. Both model evaluated one label per track over 150 tracks.

C. Task 3: Multi-Label Tagging and Fusion (MagnaTagATune)

a) The edge policy has no measurable effect: Each configuration was run with two seeds. Under $\tau = 0.3$ the two runs give AUC-PR 0.2350 and 0.2384 (mean 0.2367);

TABLE V

TASK 3: MAGNATAGATUNE TEST RESULTS. GNN ROWS ARE MEANS OVER TWO SEEDS. ALL FIGURES AT PER-TAG TUNED THRESHOLDS.

Model	Edges	Macro-F1	AUC-PR	genre	mood
Prior baseline	–	0.0706	0.0392	0.0716	0.0683
MLP, no graph	none	0.2605	0.2464	0.2843	0.2101
GNN, $\tau = 0.3$	0.97	0.2595	0.2367	0.2951	0.1842
GNN, top- $k=2$	2.00	0.2602	0.2387	0.2934	0.1900

under top- $k=2$ they give 0.2368 and 0.2406 (mean 0.2387). Mean AUC-PR therefore differs by 0.0020 between policies while the same configuration varies by up to 0.0038 against itself, and the two ranges overlap across most of their span. We take ± 0.004 as the resolution limit of these experiments throughout. Doubling similarity edges from 0.97 to 2.00 per clip and eliminating every bare-chain clip (35.0% $\rightarrow$ 0%) produced no effect distinguishable from seed noise.

b) The zero-edge rate is arithmetic, not threshold failure: At $\tau = 0.3$ the rule accepts non-adjacent pairs at 18.4% on GTZAN (1.84 edges over 10 candidate pairs) and 16.2% on MagnaTagATune (0.97 over 6) essentially the same rate. With acceptance rate $p = 0.162$ over 6 candidate pairs, the predicted zero-edge fraction is $(1 - p)^6 = 34.7\%$, against 35.0% observed. The same construction on MusicCaps yields 40.7% zero-edge clips at 0.87 similarity edges per clip: also five segments, but drawn from a 10-second excerpt in which a section rarely repeats at all. The ordering across the three corpora (GTZAN 13%, MagnaTagATune 35%, MusicCaps 41%) tracks clip duration, as the mechanism predicts.

c) Genre versus mood: The GNN exceeds the MLP on genre in *every* run (0.2928–0.2965 vs 0.2843) and falls below it on mood in *every* run (0.1794–0.1939 vs 0.2101). The two effects approximately cancel in the pooled metric. MTAT’s mood group is dominated by dynamics and tempo descriptors slow (3547), fast (2306), loud (1086), quiet (1055) with dark (193), calm (131) and mellow (96) the only frequent affect terms. We therefore describe this axis as acoustic character rather than emotion.

D. Task 3: Fusion Ablation

Three results follow, in order of how much they change the interpretation.

a) Fusion works.: The best fused model reaches AUC-PR 0.3013 against 0.2464 for the strongest single modality, a gain of +0.055 an order of magnitude larger than the ± 0.004 seed variance. Audio and text carry complementary information here, and combining them is worth doing.

b) The graph still contributes nothing: Replacing the GNN with a mean-pooling MLP *improves* AUC-PR from 0.3001 to 0.3013. The difference is within noise, so the honest statement is that the two are indistinguishable but that is the point. The no-graph result of Section VI was established without text; it survives fusion. Had we reported only crossattn against bert (0.3001 vs 0.2123), the natural reading would have been that the graph contributes +0.088, when in fact the

TABLE VI

TASK 3 FUSION ABLATION ON MAGNATAGATUNE TEST. ALL ROWS SHARE CLASSIFIER WIDTH, INITIALISATION SEED, LOSS, THRESHOLDING AND EVALUATION PROTOCOL; ONLY THE BRANCHES PRESENT AND HOW THEY COMBINE DIFFER. SINGLE-MODALITY ROWS REPEATED FROM TABLE V FOR REFERENCE. THE GENRE AND MOOD COLUMNS ARE AUC-PR WITHIN EACH TAG GROUP.

Row	Params	Macro-F1	AUC-PR	genre	mood
<i>single modality</i>					
tags_linear	19.5k	0.2559	0.2209	0.2631	0.1317
bert	105k	0.2495	0.2123	0.2582	0.1151
GNN only	172k	0.2620	0.2350	0.2642	0.1731
MLP only	105k	0.2605	0.2464	0.2732	0.1894
<i>fusion</i>					
concat	208k	0.3074	0.2945	0.3341	0.2107
crossattn	323k	0.3186	0.3001	0.3465	0.2019
mlp_bert	165k	0.3200	0.3013	0.3508	0.1964

audio features contribute that and the edges contribute nothing measurable. This is precisely what the mlp_bert control exists to expose.

c) BERT contributes nothing over the raw tags: A linear probe on the 91-dimensional instrument tag vector scores 0.2209 AUC-PR with 19.5k parameters, *above* BERT’s 0.2123 with 105k. Our MagnaTagATune text is generated from those tags by template, so BERT is effectively an expensive lookup table over a small vocabulary, and its pretraining is not exercised. This is a limitation of the text source rather than of the fusion architecture, and it is why Task 4 uses MusicCaps, whose captions are free text (Section VIII).

d) Cross-attention over early concatenation: Cross-attention exceeds early concatenation by +0.0056 AUC-PR overall, within seed noise, at a cost of 55% more parameters. With five graph nodes attending over roughly thirty text tokens there is limited structure for attention to resolve.

e) The ranking is not stable across tag groups: The last two columns of Table VI separate the fused rows by tag group, and the ordering changes depending on which group is examined. On genre the overall ranking holds, with mlp_bert highest (0.3508) and concat lowest (0.3341). On mood it reverses entirely: concat is best (0.2107), crossattn second (0.2019) and mlp_bert last (0.1964).

Two things follow. First, the graph is not uniformly inert even in fusion on mood, crossattn exceeds its no-graph counterpart by +0.0055, which inverts the direction seen without text. Second, and more importantly, all three of these differences (+0.0088, +0.0055, +0.0043) are at or below the ± 0.004 seed variance. We therefore report the three fusion strategies as indistinguishable rather than ranking them, and note that a single overall AUC-PR would have concealed the fact that the ordering is group-dependent.

E. Case Studies: Graph Paths and Caption Alignment

Table VII traces three MagnaTagATune test clips through the cross-attention model, reporting for each the graph’s similarity edges, the caption tokens the graph readout attends to, and the predicted against true tags. All three clips yield five

TABLE VII

THREE TEST CLIPS THROUGH THE CROSS-ATTENTION FUSION MODEL. “SIM. EDGE” IS THE NON-ADJACENT SEGMENT PAIR RETAINED BY THE $\tau = 0.3$ RULE, WITH ITS TRACK-CENTRED COSINE. ATTENTION WEIGHTS ARE OVER CAPTION TOKENS.

Clip	Sim. edge	Caption, attention and outcome
29 <i>Bach</i>	2 $\leftrightarrow$ 4 0.662	“...featuring violin.” violin 0.996, featuring 0.002 true classical; predicted classical
33 <i>harpsi.</i>	2 $\leftrightarrow$ 4 0.529	“...harpsichord and harpsicord.” harpsichord 0.9998 true classical, loud, fast; predicted classical only
133	<i>none</i>	“A music clip.” clip 0.42, ‘.’ 0.22, a 0.16, [CLS] 0.10 true new age; predicted nothing

segments and the same four-edge temporal chain, so the rows differ only in what the model finds.

a) Attention is sharp and correct when there is content:

On clips 29 and 33 the graph readout places essentially all of its attention mass on the single instrument token 0.996 on violin and 0.9998 on harpsichord and almost none on the template scaffolding (a, music, clip, featuring). The cross-attention layer has learned to ignore the parts of the sentence that are constant across the corpus. Both clips are correctly assigned classical.

b) Clip 33 shows the genre–mood asymmetry inside a single example: The model recovers classical but misses loud and fast. This is the expected consequence of the vocabulary partition: harpsichord is highly informative about genre and says nothing whatever about dynamics or tempo, so those two tags have to be inferred from the audio branch alone and a five-node graph of pooled MFCC statistics does not supply them. The aggregate result of Section VI, where mood scores consistently below genre, is this pattern repeated across the test set.

The caption also illustrates the tag redundancy discussed in Section III: harpsichord and its misspelling harpsicord are separate tags in the MagnaTagATune vocabulary and both fire on this clip, so the generated sentence names the instrument twice. Because instrument tags are inputs rather than targets, this costs nothing.

c) Clip 133 is the degenerate case on both branches.:

The clip has no instrument tags, so its caption is the bare fallback “A music clip.”; and its graph has *no* similarity edge, leaving a bare temporal chain the condition affecting 35.0% of clips at $\tau = 0.3$ (Section VI). With nothing to attend to, the attention distribution flattens across function words and punctuation (0.42/0.22/0.16/0.10) rather than concentrating anywhere, and the model predicts no tags at all, missing the true label new age. It is worth noting that the model does not fabricate confidence in this situation: a near-uniform attention distribution is the correct response to an input that carries no information.

TABLE VIII

TASK 4: CAPTION–AUDIO RETRIEVAL OVER AN 796-CLIP TEST GALLERY. CAPTIONS ARE 100% UNIQUE PER CLIP.

	R@1	R@5	R@10
Random embeddings	0.0025	0.0075	0.0113
Caption $\rightarrow$ audio	0.0126	0.0766	0.1382
Audio $\rightarrow$ caption	0.0126	0.0678	0.1332
Lift over chance	10.0 $\times$	12.2 $\times$	11.0 $\times$

TABLE IX

TASK 4: ZERO-SHOT TAGGING AGAINST A SUPERVISED REFERENCE TRAINED ON THE SAME MUSICCAPS TAGS (34 GENRE + 21 MOOD) AND SPLITS. SUPERVISED FIGURES AT A FIXED 0.5 THRESHOLD; PER-TAG TUNING WAS MILDLY HARMFUL HERE, SEE TEXT.

Model	Supervision	Macro-F1	AUC-PR	genre	mood
Prior baseline	–	0.0850	0.0462	0.0793	0.0942
Contrastive, zero-shot	none	0.0919	0.0535	0.0852	0.1028
Supervised GNN	3,709	0.1783	0.1502	0.1857	0.1662

F. Threshold Behaviour on a Single Clip

The GNN-only model’s ranked output for clip 29 shows how the per-tag thresholds behave. The true tag classical is ranked first at probability 0.841 against its threshold of 0.72, followed by slow (0.623 vs 0.60, fires), soft (0.578 vs 0.60, does not), ambient (0.490 vs 0.74) and quiet (0.465 vs 0.66).

Across the full tag set the thresholds span 0.33 to 0.93. A single global cut at 0.5 would fire soft and fast here and suppress nothing, which is why per-tag tuning is applied despite its modest average benefit.

More instructive is medieval, which fires despite not appearing among the ten highest probabilities: its tuned threshold sits below 0.224, so a low score suffices. This is the failure mode of per-tag thresholding on rare tags a threshold fitted to a handful of validation positives can be driven low enough that the tag fires on weak evidence. It is the same mechanism behind the negative tuning result on MusicCaps, where fitting 55 thresholds on 794 validation clips reduced Macro-F1 by 0.010.

G. Task 4: Cross-Modal Alignment (MusicCaps)

Table VIII reports retrieval in both directions against a random-embedding baseline, and Table IX the zero-shot tagging result against a supervised reference trained on the same data. Figure 4 shows the learned shared space: structure is markedly weaker than in Figure 3, with most clips falling in the grey *other* category and no tag forming a compact region. That is the geometric counterpart of R@1 = 0.0126 the space encodes musical character well enough for 10–12 $\times$ chance retrieval, but not finely enough to separate individual clips.

The contrastive model recovers 36% of supervised AUC-PR (0.0535 vs 0.1502) with no tag supervision at any stage, and exceeds the label-prior baseline by 1.16 $\times$. Notably the ordering inverts relative to the supervised model: zero-shot performs *better* on mood than genre (0.1028 vs 0.0852)

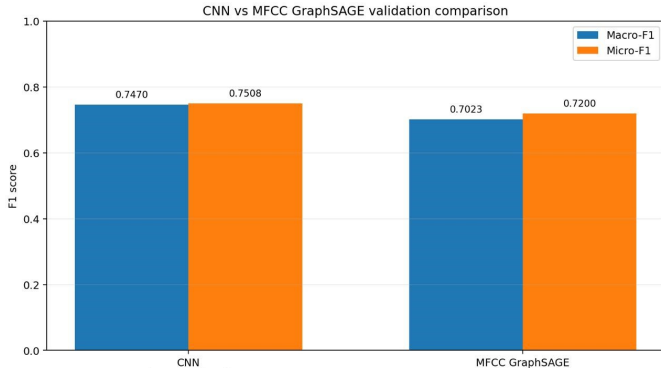


Fig. 2. Task 2: Comparison of CNN and GNN validation scores.

whereas supervision favours genre (0.1857 vs 0.1662). MusicCaps captions describe affect in ordinary language that the text encoder aligns to readily, whereas genre terms are rarer and more categorical.

Three of the ten examples in Table X are worth examining in detail, chosen to span the behaviour rather than to flatter it.

The failure mode is informative. In the first example the model is asked for an electronica instrumental with fast synthesiser articulations and digital drums, and returns clips tagged electronic, techno, energetic and electronic, hip hop the wrong clips, but the right region of the space. The same holds elsewhere: a query for a tropical steel-drum piece returns clips tagged happy and energetic, and a query for a mellow clean electric guitar with delay returns ambient, cinematic, meditative. The embedding has learned musical character; what it has not learned is to distinguish one clip from several hundred others sharing that character, which is precisely what R@1 measures.

The second example is the one exact hit, and it is revealing that it succeeds on a *recording-quality* description low quality, mono, plastic percussion, probably recorded on a phone rather than on musical content. Production artefacts are unusual enough to be nearly unique, whereas “upbeat pop with male vocals” describes a large fraction of the corpus.

The third is a genuine failure: a spoken narration about French horn repair over an orchestral brass section, retrieving two untagged clips and one tagged children’s music. Speech-over-music sits outside what a five-node MFCC summary can represent.

Table X lists all ten required examples in compact form. The correct clip appears in the top three for one of the ten, consistent with the measured R@10 of 0.1382 over a 796-clip gallery, and the retrieved clips are plausible in character throughout: electronic queries return electronic/techno clips, relaxed queries return calm/happy clips.

VII. DISCUSSION

A. Why the Graph Does Not Help

The readout is `global_mean_pool`. With five nodes, a four-edge temporal chain, and one to two similarity edges, message passing primarily acts as a learned smoothing operation before

the node representations are averaged. As a result, the GNN may be learning information that is largely discarded by the final mean pooling. This is consistent with the no-graph MLP, which operates directly on the mean of the same node features, uses fewer parameters, and converged more stably: its validation loss reached 0.3545 over 75 epochs, compared with 0.3587 for the GNN over 12–22 epochs.

This finding should be interpreted specifically for the five-node segment graphs derived from short excerpts, rather than as a general statement about graph neural networks for music. The graph structure also differs from Task 2: Task 2 uses six nodes from full 30-second tracks, with ten candidate non-adjacent pairs, whereas the MagnaTagATune graphs contain five nodes and six candidate non-adjacent pairs.

B. The Genre–Mood Asymmetry Is Visible in the Embedding

Figure 3 provides a more detailed view of the MagnaTagATune embedding by colouring the same representation by genre and mood. Under genre, classical, opera, and rock occupy compact and largely disjoint regions, while ambient and the electronic/techno family form distinct clusters. Mood also shows structure, but largely within the same regions. The area occupied by rock, for example, is where loud, heavy, and fast concentrate, while the region occupied by classical contains soft, quiet, and slow.

This suggests that the embedding has one dominant organising axis rather than separate axes for genre and mood. It is primarily organised by timbral family, with mood tags becoming more distinguishable when they correlate with timbre. The mood group in MagnaTagATune is dominated by dynamics and tempo descriptors, which correlate strongly with this structure and are therefore predicted well above chance. In contrast, affect-related tags that do not follow these patterns remain more scattered.

This interpretation is consistent with the numerical results in Section VI. The GNN exceeds the no-graph MLP on genre in every run but falls below it on mood in every run. Segment-to-segment repetition is a property of musical form, and musical form aligns more closely with genre than with loudness. Consequently, the information introduced by message passing reinforces the axis already captured by the embedding, while adding little information for the mood-related structure that is not aligned with it.

C. Calibration Dominates Thresholding

Per-tag threshold tuning improves Macro-F1 by approximately +0.012 across the MagnaTagATune runs. However, the larger effect comes from calibration before thresholding. An earlier configuration without the `pos_weight` cap produced mean precision of 0.183 and mean recall of 0.441, indicating systematic over-prediction that threshold tuning could only partially correct. Capping `pos_weight` at 10 improved the upstream calibration and consequently left less error for threshold tuning to recover.

The effect was different on MusicCaps, where threshold tuning was mildly harmful (−0.0101). This is consistent

TABLE X

ALL TEN QUALITATIVE CAPTION $\rightarrow$ AUDIO RETRIEVAL EXAMPLES ON THE MUSICCAPS TEST SPLIT. “TOP-1 TAGS” ARE THE GROUND-TRUTH TAGS OF THE HIGHEST-SCORING RETRIEVED CLIP; “Hit” GIVES THE RANK AT WHICH THE CLIP THE QUERY ACTUALLY DESCRIBES APPEARS, OR – IF IT IS ABSENT FROM THE TOP THREE. RETRIEVED CLIPS ARE CONSISTENTLY PLAUSIBLE IN CHARACTER EVEN WHEN THE EXACT CLIP IS MISSED.

#	Query caption (truncated)	Top-1 tags	Sim.	Hit
1	A male vocalist sings this upbeat Arabic pop. The tempo is fas...	indie, pop, fun	0.599	–
2	This is a tropical, Caribbean style song which has a melody th...	pop, energetic, happy	0.539	–
3	A male vocalist sings this upbeat pop song. The tempo is mediu...	dance, electronic, hip hop	0.567	–
4	This is a Hindi film music piece. This is an instrumental piec...	groove, fun, happy	0.492	–
5	The low quality video features a cover of a kids song and it c...	love, pop, emotional	0.612	–
6	The low quality recording features plastic percussion hits. Th...	calm, meditative	0.617	3
7	This is a gear showcase jam. The only instrument that is being...	ambient, cinematic, meditative	0.570	–
8	A man narrates a repair technique for a French horn. The song ...	<i>untagged</i>	0.541	–
9	The Jazz song features a wide brass section playing over soft ...	romantic	0.553	–
10	This music is an Electronica instrumental. The tempo is fast w...	electronic, groove, techno	0.547	–

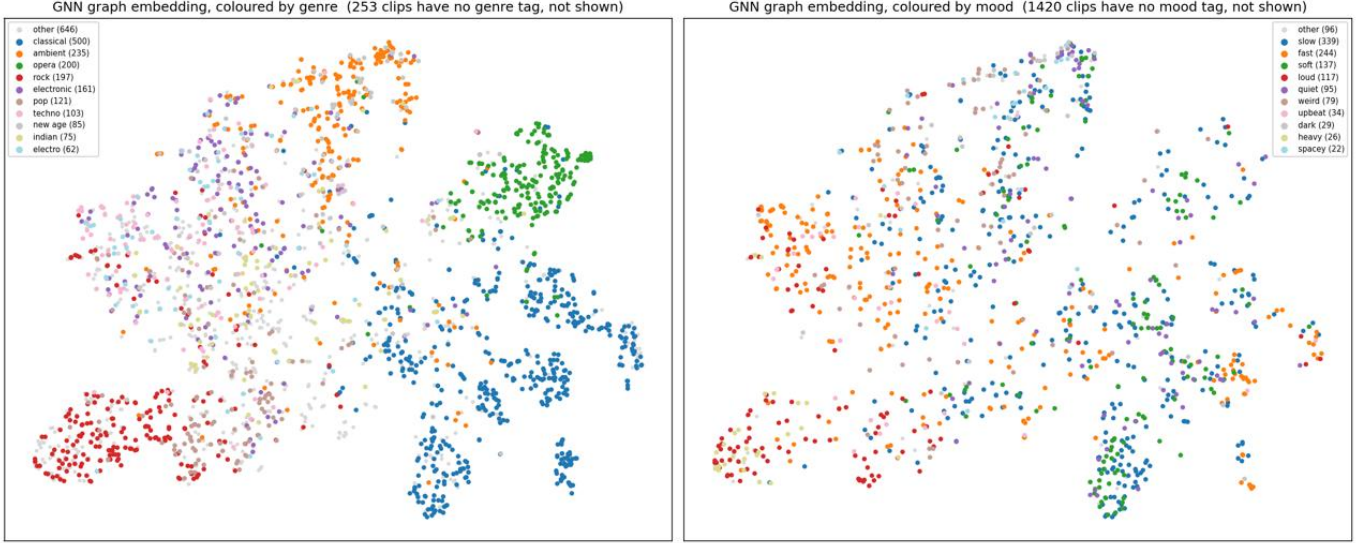


Fig. 3. Task 3: t-SNE of the GNN graph embedding on the MagnaTagATune test split, coloured by genre (left) and by mood (right). Each clip takes the colour of its *rarest* positive tag, since the most frequent tag would otherwise dominate; clips with no tag in that group are omitted and counted in the title. Genre forms compact, well-separated regions: classical occupies the right half in several dense clusters, opera a single tight mass, rock the lower left, and ambient the upper centre. The mood panel is not unstructured, but its structure is *inherited*: fast and loud occupy the same lower-left region as rock, and soft and slow the same right-hand region as classical. The mood axis is largely a projection of the timbral organisation rather than an independent one, which is what Section VI measures as the lower mood scores.

with fitting 53 thresholds using only 794 validation clips, making the per-tag thresholds more susceptible to validation-set variation.

VIII. LIMITATIONS AND FUTURE WORK

Several limitations point to clear directions for future work. First, the audio representation remains a major constraint. Each clip is summarized by 26 pooled MFCC statistics, rather than the full mel spectrograms used by published music taggers.

Second, each configuration was evaluated with only two random seeds. As a result, differences below approximately 0.004 AUC-PR cannot be reliably resolved at the current sample size. Future experiments should use more random seeds to provide more robust estimates of variance and statistical significance.

Third, the MusicCaps zero-shot evaluation has an inherent caption-tag circularity. The curated tags are derived from aspect_list and were written by the same annotator as the caption. Thus, “zero-shot” in our setting indicates the absence

of tag supervision during model training, rather than evaluation against a completely independent label set. Future evaluation with independently annotated labels would provide a stronger test of zero-shot generalisation.

Finally, zero-shot tagging uses a single prompt template, whereas CLIP-style systems commonly ensemble multiple prompts. Exploring prompt ensembles or alternative prompt formulations could therefore provide a stronger zero-shot baseline.

IX. CONCLUSION

We built a GNN-BERT system for multi-label music context understanding across three corpora and evaluated it against controlled baselines at every level. Text alone recovers tags well (0.6462 AUC-PR on MusicCaps captions), but largely because the tags were derived from those captions. Audio alone, represented as 26 pooled MFCC statistics over five or six segments, reaches 0.2464 on MagnaTagATune and 0.1502

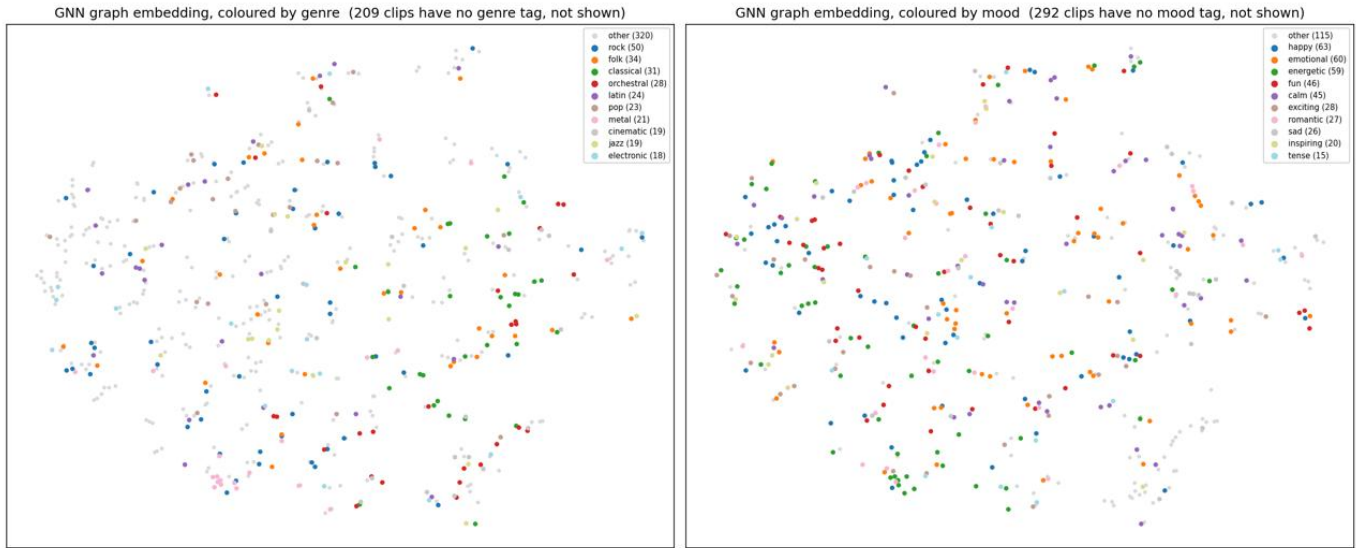


Fig. 4. Task 4: t-SNE of the shared audio–text embedding on the MusicCaps test split, coloured by genre (left) and mood (right). Structure is markedly weaker than in Fig. 3: most clips fall in the grey *other* category and no tag forms a compact region. This is the geometric counterpart of $R@1 = 0.0126$ the space encodes musical character well enough for 10–12 $\times$ chance retrieval, but not finely enough to separate individual clips.

on MusicCaps. Fusing the two is clearly worthwhile: 0.3013 AUC-PR, an improvement of +0.055 over the best single modality and an order of magnitude beyond the ± 0.004 seed variance we measured.

The graph structure, however, is not what makes fusion work. An MLP given the same features with every edge removed matches the GNN both alone (0.2464 vs 0.2387) and in fusion with BERT (0.3013 vs 0.3001), and doubling the number of similarity edges changes nothing beyond seed noise. The mechanism is that with five nodes, a mean-pooling readout and a 26-dimensional node feature, message passing largely smooths features that are then averaged away. The one consistent exception is informative: the GNN is better on genre in every run and worse on mood in every run, matching the composition of the tag vocabularies.

We report this because the alternative would have been easy and misleading. Comparing our cross-attention model against BERT alone shows a gain of +0.088 AUC-PR, which reads naturally as evidence that the graph helps; the control shows that the audio features are responsible and the edges are not. Segment graphs may well earn their place at a different scale longer clips, more nodes, a readout that preserves structure, or spectrogram-level graphs as in recent audio GNN work but that claim needs the control to be credible, and on this data it does not hold.

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